Demodulate, Then Attend: Exposing Amplitude Dynamics for Compact EEG Models

Anonymous CVPR submission

Abstract

Modern electroencephalography (EEG) models increasingly rely on larger networks and more pretraining. We revisit a simpler choice: the signal representation on which learning operates. We advocate demodulation before attention: extract each band's amplitude envelope and pair it with the corresponding waveform, exposing both amplitude dynamics and oscillatory morphology. Scattering theory motivates this separation and explains why temporal averaging can discard useful modulation structure. A simple tokenizer applies a learnable filterbank and embeds matched local waveform and envelope sequences into an electrode–band–time grid. A compact axial Transformer reads this representation. With the reader and training recipe fixed, this representation improves TUEV Cohen's κ from 0.576 to 0.711, an absolute gain of 0.135, while using fewer parameters; the same construction achieves 0.699 PR-AUC on CHB-MIT with 1.62 million parameters and no pretraining. These results motivate a representation-first approach to compact EEG models: make useful signal structure available in the tokens, then learn how it is organized across time, electrodes, and frequency.

1. Introduction

EEG modeling has progressed from compact convolutional networks to Transformers pretrained on large collections of recordings [5, 6, 13, 15]. Larger models promise broader transfer across clinical tasks, subjects, and acquisition systems. Yet scale alone does not determine downstream performance: recent benchmarking finds that supervised specialists remain competitive with pretrained EEG models [7]. This observation suggests an input-side question. Before asking how much capacity an encoder needs, what signal structure should its tokens already expose?

Clinical EEG interpretation involves both the shape of a transient and the organization of ongoing activity. Morphology, recurrence, evolution, and spatial distribution all enter standardized descriptions [4]. A sharp discharge and a smooth oscillation differ in shape; sustained activity and a sequence of bursts differ in amplitude dynamics. These distinctions motivate two complementary views of each frequency band: its signed waveform and its amplitude envelope. The first retains oscillatory detail, and the second exposes when activity strengthens or weakens. A representation should make their correspondence available at the same electrode and time.

Linear patch embedding leaves this separation implicit. An envelope is present in the recorded samples, but extracting it requires a band-selective transform and a nonlinear modulus. A linear map of a raw patch cannot perform that computation on its own. The subsequent network can learn nonlinear features, but doing so becomes part of its supervised learning problem. Our hypothesis is that a compact encoder benefits when these amplitude dynamics are supplied before attention, alongside waveform content that amplitude alone cannot specify.

Analytic demodulation gives this hypothesis a concrete foundation. For an amplitude-modulated oscillation, the analytic modulus removes the rapid carrier and exposes its amplitude variation. Scattering theory builds representations by composing bandpass transforms with modulus operations and shows how temporal averaging can discard modulation structure [1]. This distinction is central here: *extracting amplitude is not the same as retaining amplitude dynamics*. Two intervals can have the same average band energy and different burst patterns. A sequence of envelope values retains that difference; a single pooled energy does not.

The resulting design is simple. Demodulate each band before patch embedding, preserve a short envelope trajectory, and pair it with the waveform from that same band and interval. Each token then describes both what a local oscillation looks like and how its amplitude changes. Keeping electrodes, bands, and time intervals as separate indices gives the encoder an explicit grid on which to learn interactions. An ordinary axial Transformer reads this grid; no separate coupling estimator, phase rotation, or reconstruction objective is required.

This construction builds on, rather than replaces, earlier spectral frontends. CBraMod adds patchwise spectral and convolutional waveform features [15]. TFM-Tokenizer learns discrete motifs from waveform and time–frequency

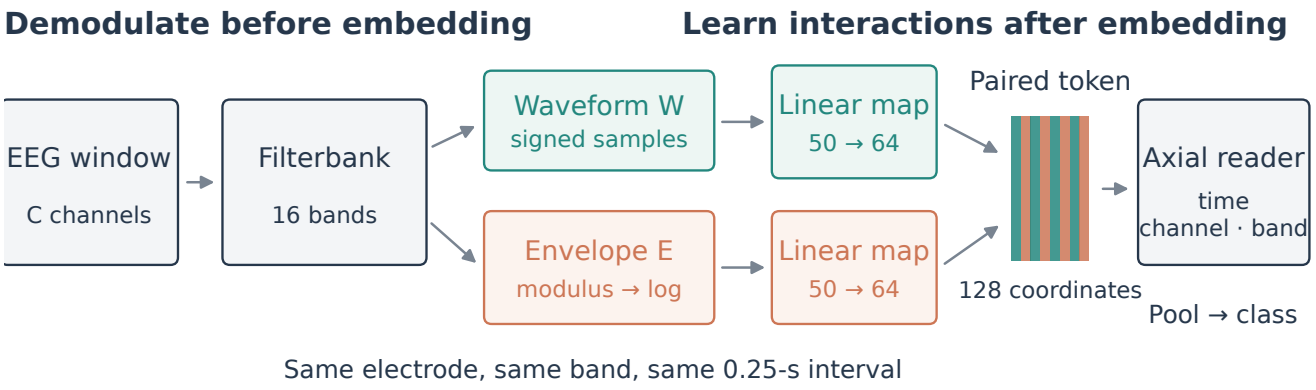


Figure 1. **Demodulate, then attend.** A learned filterbank produces band-limited waveforms. The analytic modulus supplies a corresponding amplitude envelope. Shared projections embed both 50-sample sequences into one token, retaining their electrode, band, and time indices. Six axial Transformer blocks learn interactions on this grid. The analytic transform is applied to the full input window before patching.

representations [10]. The Morlet Spectral Transformer preserves time-resolved wavelet amplitudes [11]. Our focus is the *local pairing before attention*: within each band, a token receives the envelope trajectory together with its corresponding waveform. Content controls with a fixed reader make this representation choice a testable object, rather than attributing an entire architecture's performance to its frontend.

The unrotated reference reaches $\kappa = 0.711$ on TUEV and PR-AUC 0.699 on CHB-MIT using 1.62 million parameters. On TUEV, replacing it with raw patches projected into the same number of learned rows reduces κ to 0.576 under the same reader, split, and training recipe. The proposed frontend improves F1 in all six event categories. A separate frozen-token readout recovers both local content views, checking that the tokenizer actually preserves what the design intends to expose.

Our contribution is a representation choice and a way to examine it: demodulate before embedding, pair corresponding waveform and amplitude trajectories, and hold the reader fixed when testing their predictive roles. This shifts the emphasis from adding machinery to deciding which signal computations should precede learned attention. The broader question is how much of a compact model's effectiveness comes from the information made directly usable at its input.

2. Related Work

2.1. EEG pretraining and tokenization

EEG foundation models address transfer across subjects, tasks, and recording systems. BIOT organizes channelwise signal segments into a common sequence [16]. LaBraM learns discrete neural codes through spectrum prediction [5], and EEGPT uses masked representation learning with alignment [14]. CBraMod separates spatial and temporal interactions in a criss-cross Transformer [15]; REVE extends large-scale pretraining to heterogeneous acquisition setups [2]. These methods combine input representations, learning objectives, and encoder designs to obtain transferable features.

TFM-Tokenizer instead makes the frontend a primary object of study: waveform and spectral paths produce a vocabulary of time–frequency motifs that can serve different encoders [10]. Our construction uses continuous, explicitly demodulated band content. It does not require a learned codebook or a separate tokenizer pretraining stage. The question is which local signal views should remain available inside each token before the encoder mixes information across locations.

2.2. Amplitude-based EEG representations

Explicit amplitude information is well established in EEG decoding. The shallow network of Schirrmeister et al. uses squaring, pooling, and logarithmic compression to extract power-related features [13]. FBCNet combines a filterbank, spatial filtering, and variance aggregation [8]. EEGNet provides another compact architecture through factorized temporal and spatial filtering [6]. These precedents show the value of building signal operations into small learned models.

Recent frontends also preserve spectral or amplitude variation. CBraMod projects a one-second patch spectrum and combines it with waveform features [15]. The Morlet Spectral Transformer uses wavelet modulus, temporal pooling, and frequency-specific spatial projections [11]. Our distinction is narrower than introducing amplitude: a token embeds a sequence of envelope samples and the corresponding signed waveform from one band, while electrodes remain a token axis. This makes local content and the gran-

ularity of its representation explicit experimental variables.

2.3. Demodulation before learned processing

The deep scattering spectrum extracts subband envelopes with analytic wavelets and modulus operations, then recovers modulation structure that would be lost through averaging [1]. We adopt the demodulation principle while passing local trajectories to learned attention. The implementation uses trainable sinc filters [12], rather than a fixed scattering cascade. Its contribution is the token interface, not a new demodulation theorem.

These signal-processing precedents suggest an input-side study of compact models: explicitly expose useful structure and measure what a fixed reader can learn from it.

3. Method

Let $x_c[t]$ be the normalized EEG signal at channel c. We construct one token per channel c, band b, and time interval p. The method has three steps: demodulate the bands, embed paired local content, and read the resulting grid (Fig. 1).

3.1. Bandwise demodulation

A shared bank of B learnable bandpass filters produces signed waveforms $w_{c,b}$ and their analytic envelopes:

$$\begin{aligned} w_{c,b} &= g_b * x_c, \\ e_{c,b} &= \left| w_{c,b} + \mathrm{i}\mathcal{H}[w_{c,b}] \right|. \end{aligned} \tag{1}$$

where $\mathcal{H}$ is the Hilbert transform. Each sinc filter g_b is parameterized by two cutoffs [12]. We use 16 filters, initialized with logarithmically spaced edges from 0.5 to 75 Hz, and a 201-sample impulse response at 200 Hz. The cutoffs are learned during supervised training.

For a narrowband amplitude-modulated component $w(t) = A(t)\cos\phi(t)$ with suitable separation between modulation and carrier frequencies, the analytic envelope approximates $A(t)$. Thus the modulus exposes amplitude dynamics that would otherwise need to be derived from oscillating samples. With general EEG, it is the well-defined amplitude of the filtered analytic signal; it need not identify a single physiological oscillator. We retain the signed waveform in parallel because taking a modulus removes its phase and polarity information.

Filtering and the analytic transform operate on the entire input window. Only afterward do we divide the two sequences into tokens. A 0.25-s token therefore summarizes a local part of an envelope estimated with context beyond that token. The token interval is not the filter's support and does not impose a 0.25-s Hilbert-estimation window.

3.2. Pair local waveform and envelope trajectories

For interval p of length L, let $\mathbf{w}_{c,b,p} \in \mathbb{R}^L$ and $\mathbf{e}_{c,b,p} \in \mathbb{R}^L$ contain corresponding waveform and envelope samples. We form

$$\mathbf{h}_{c,b,p} = \left[P_E \log(1 + \mathbf{e}_{c,b,p}) + \mathbf{b}_E \,;\, P_W \mathbf{w}_{c,b,p} \right], \tag{2}$$

with $P_E, P_W \in \mathbb{R}^{(d/2)\times L}$. Both projections are shared across channels, bands, and intervals. The implementation interleaves the two sets of coordinates, which is a fixed permutation of the concatenation in Eq. (2). We use $L = 50$ and $d = 128$. The logarithm compresses envelope dynamic range without averaging its trajectory.

The pairing is deliberately local. Each token places waveform and amplitude dynamics from the same signal component in a common learned representation. The encoder can then model their relationship without first matching independently aggregated views. We preserve all 50 input positions through separate projections rather than substituting a single power or variance statistic. There is no phase rotation or additional learned fusion module at this stage.

This construction also separates *available information* from *explicit computation*. A raw-patch projection can preserve all its input samples when its matrix has sufficient rank. It nevertheless does not output an amplitude envelope: the modulus is nonlinear. Our design supplies this computation before attention and retains the waveform from which it was derived. It introduces a representational bias, not an additional measurement of the underlying EEG.

3.3. A compact reader for the token grid

Tokens form a $C \times B \times P$ grid. Channel embeddings derived from electrode coordinates and learned band-index embeddings identify the two non-temporal axes. Each pre-normalized Transformer block applies self-attention along time, channels, and bands in sequence, followed by a feed-forward network. Time attention uses rotary positions. Each axis uses ordinary four-head attention; the reference has six blocks, width 128, and a feed-forward expansion of two.

The axes follow the representation: time relates intervals, channels relate recording locations, and bands relate signal components. Factorization permits this grid to be read without global attention over all its tokens. Global mean pooling, layer normalization, and a linear classifier predict the window label. The complete reference contains approximately 1.62 million parameters. We keep this reader fixed in the content experiments; it is not a separate proposed attention mechanism.

3.4. Checking the token's content

Before attributing a classification result to preserved content, we check that the two intended views can be recovered from the token. We freeze the trained frontend and fit separate linear readouts to waveform and log-envelope samples,

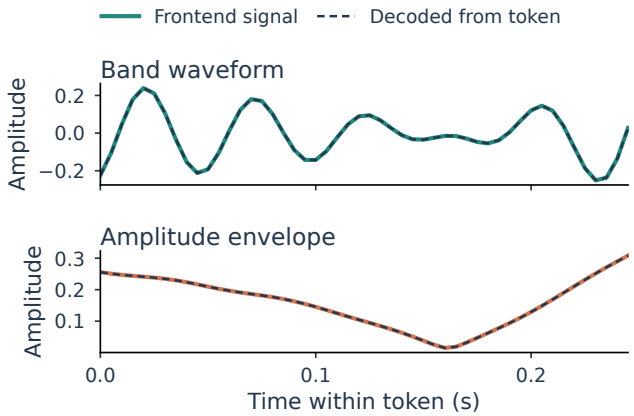


Figure 2. **What remains in a token?** A linear readout of frozen TUEV tokens recovers the waveform and envelope of a held-out band segment. Curves overlap at this scale. This checks local content preservation before attention; classification controls test utility.

Table 1. **Content recovery before attention.** Held-out normalized RMSE of linear readouts fitted to frozen tokens. Targets are the frontend's waveform and log-envelope samples.

Target	TUEV	CHB-MIT
Waveform	3.21×10^{-4}	1.33×10^{-4}
Log-envelope	4.13×10^{-7}	5.38×10^{-7}

using training windows only. Readout regularization is selected on validation windows; reconstruction is measured on held-out test windows. No classification labels enter this check, and no classifier weights are updated.

Both content views are recovered with small error (Tab. 1 and Fig. 2). The two 64×50 projection matrices have full column rank in both trained models, explaining why linear recovery is possible. This is a functional check of the representation, not evidence that every retained coordinate is needed for classification. In particular, reconstruction does not establish whether a waveform-only or envelope-only token would perform equally well. Those questions require the content controls in Sec. 5.2.

4. Experimental Setup

Tasks and data. We study six-class event classification on TUEV [9] and binary seizure-window classification on CHB-MIT [3]. Both use 16 bipolar channels at 200 Hz. TUEV contains 5-s event-centered windows, with subject-separated training and validation sets drawn from the official training partition and the official evaluation subjects reserved for testing. CHB-MIT uses 10-s windows and case-separated partitions. Its evaluated protocol includes additional seizure-centered windows in each partition; the test set has 17,386 windows, including 276 positives. All within-dataset comparisons must use these same windows. Appendix A gives normalization, partition counts, and the exact sampling protocol.

Training. Models are trained from scratch with AdamW, a learning rate of 10^{-4}, weight decay 0.05, cosine scheduling, batch size 32, and a maximum of 20 epochs. The configured dropout is 0.35. TUEV uses cross-entropy with label smoothing 0.1; CHB-MIT uses focal loss with $\alpha = 0.25$ and $\gamma = 2$. All content arms share the same data augmentation, optimizer, reader, and checkpoint-selection rule. Appendix B details the transformations and validation schedule.

Evaluation. We select checkpoints by validation κ on TUEV and validation PR-AUC on CHB-MIT. The held-out test set is used for reporting. TUEV reports κ, balanced accuracy, and support-weighted F1. CHB-MIT reports trapezoidal PR-AUC, AUROC, and balanced accuracy at threshold 0.5; we additionally give average precision. The completed comparisons use the fixed seed 2, and tables leave unavailable results empty. Baseline scores from incompatible montages or window sets are not substituted into these comparisons.

5. Results

5.1. Clinical performance

The reference achieves $\kappa = 0.711$, balanced accuracy 0.634, and weighted F1 0.849 on TUEV (Tab. 2). On CHB-MIT it achieves PR-AUC 0.699 and AUROC 0.918. Average precision is 0.702, close to the trapezoidal PR-AUC. These results are obtained with supervised training alone and a compact common reader.

The directly controlled TUEV comparison changes the input representation. Raw-patch tokenization maps each 50-sample channel segment into 16 learned rows of width 128. It therefore supplies the same grid size and attention structure as the bandwise frontend, without filtering or demodulation. It obtains $\kappa = 0.576$. Exposing bandwise waveform and amplitude trajectories raises κ by 0.135, balanced accuracy by 0.081, and weighted F1 by 0.067. The raw-row model actually has more parameters (1.72M versus 1.62M), so the gain is not obtained by increasing parameter count. This comparison establishes the value of the complete input construction under a fixed reader; the next controls separate its content.

5.2. Which content contributes?

A waveform-only arm tests whether filtering and the band grid suffice. An envelope-only arm tests how much classification needs oscillatory shape beyond amplitude dynam-

Table 2. **Clinical evaluation under a common data protocol.** Both tasks use 16 bipolar channels. The raw-patch row uses the same three-axis reader and training recipe as our tokenizer. Empty baseline cells are reserved for protocol-matched evaluations; they do not denote zero performance. Bold marks the best available value per metric. Parameter counts are for the TUEV classifier.

		TUEV			CHB-MIT		
Model	Params (M)	$\kappa\uparrow$	Bal. acc.↑	wF1↑	PR-AUC↑	AUROC↑	Bal. acc.↑
BIOT [16]							
LaBraM [5]							
EEGPT [14]							
CBraMod [15]							
REVE [2]							
TFM-Tokenizer [10]							
Raw-patch rows + axial reader	1.72	0.576	0.552	0.782			
Demodulation-first (ours)	1.62	**0.711**	**0.634**	**0.849**	**0.699**	**0.918**	**0.630**

Table 3. **Content controls with a fixed reader.** All bandwise variants keep 16 bands, two 64-coordinate projection blocks, and the same training recipe. Repeating one view in both blocks preserves projection dimensions and parameter count. Unfinished cells remain empty.

Token input	TUEV κ	CHB-MIT PR-AUC
Raw learned rows	0.576	
Waveform only		
Envelope only		
Envelope + waveform	**0.711**	**0.699**

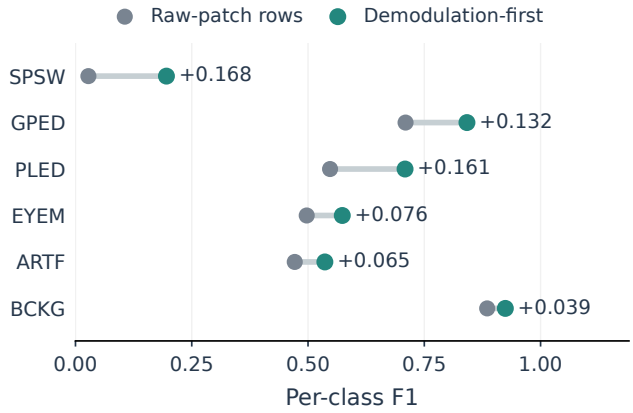


Figure 3. **TUEV gains across event categories.** Paired markers compare held-out F1 for raw learned rows and demodulation-first tokens with the same reader. Text gives the absolute difference. All six categories improve; the figure does not attribute these gains to the envelope alone.

ics. Both retain the reference's two projection blocks and reader; they replace only the signal supplied to each block (Tab. 3). This avoids conflating removal of a view with a smaller token or a smaller encoder.

The comparisons answer two distinct questions. Envelope versus waveform isolates the type of band content made explicit. Pairing both versus either alone tests complementarity at the token interface. The completed raw-row comparison supports the whole construction, while these bandwise arms determine which part of its signal content accounts for the gain.

5.3. Where does event recognition improve?

The aggregate gain extends beyond the majority background category (Fig. 3). GPED F1 increases from 0.710 to 0.842, and PLED F1 from 0.547 to 0.709. F1 also improves for spike/sharp-wave events, eye movements, artifacts, and background. These results connect the overall score to both recurrent-discharge and other event categories, rather than supporting an explanation confined to a single label type.

The class breakdown also shows what the aggregate metrics hide. Spike/sharp-wave F1 rises from 0.028 to 0.196, but recall remains 0.109. For eye movements, F1 improves despite a decline in recall, because precision rises. Table 5 in the appendix reports precision and recall alongside F1. This distinguishes improved category-level discrimination from uniform improvements in every operating characteristic.

5.4. What the representation makes explicit

The recovery check and classification comparison play different roles. The former shows that trained tokens retain the intended local trajectories before attention. The latter shows that replacing raw learned rows with this bandwise representation improves TUEV performance at fixed reader capacity. Together they connect the implemented content interface to an observed predictive gain. They do not require a new attention rule or an auxiliary objective.

Matched label-budget and capacity experiments can further test whether this representation reduces what must be learned from supervision.

6. Discussion and Conclusion

Scope. The current paired comparison fixes one reader, one seed, and one training recipe. It establishes a frontend effect under those conditions, not a universal advantage over pretrained models or a proof of reduced sample complexity. CHB-MIT results describe the specified enriched window protocol rather than continuous-recording alarm performance. The filterbank and full-window analytic transform can introduce boundary and overlap effects; their implementation is documented in the appendix.

Conclusion. We present a simple way to construct EEG tokens: demodulate before embedding and retain the corresponding waveform alongside its amplitude trajectory. A standard axial reader turns these local descriptions into clinical predictions. The observed TUEV gain shows that input representation can materially change what a compact model achieves without increasing its capacity. The resulting research direction is concrete: identify useful signal computations, expose their outputs without discarding complementary detail, and let attention learn the relationships that remain.

LLM Usage Statement

OpenAI Codex assisted with literature retrieval, manuscript drafting and editing, code and result-record inspection, figure preparation, and LaTeX compilation. It also assisted with numerical consistency checks and the organization of the experimental evidence.

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Table 4. **Evaluated window counts.** Counts include the specified CHB-MIT seizure-window enrichment.

Dataset	Train	Validation	Test
TUEV	68,445	15,487	29,421
CHB-MIT	316,205	21,184	17,386

A. Data and Evaluation Protocol

TUEV. The evaluated TUEV v2.0.1 cache contains five-second windows around annotated events. Signals are resampled to 200 Hz, bandpass filtered from 0.3 to 75 Hz, and notch filtered at 60 Hz. Values in microvolts are divided by 100 before storage. The official training subjects are sorted and split 80:20 into training and validation partitions; official evaluation subjects form the test partition. There are 232, 58, and 80 subjects, respectively. We do not apply a second scaling operation in the model loader.

CHB-MIT. The evaluated CHB-MIT v1.0.0 cache resamples the recordings from 256 to 200 Hz and extracts non-overlapping ten-second windows, dropping incomplete tails. Any overlap with an annotated seizure gives a positive label. Additional windows are extracted around seizure intervals with one-second pre/post margins and a five-second stride; these are labeled positive. This enrichment applies to training, validation, and test partitions, adding 1,781, 90, and 162 windows, respectively. It is part of the evaluated protocol, not training augmentation. A natural-grid-only evaluation would be a different test set.

Each window and channel is divided by its own 95th percentile absolute amplitude plus 10^{-8}. Thus amplitude in this experiment refers to variation within this normalization, not calibrated clinical voltage. Training uses cases chb01–chb19 and chb21; validation uses chb20 and chb22; testing uses chb23 and chb24. The repeated-subject recordings chb01 and chb21 both remain in training. The test set has 17,110 negative and 276 positive windows (1.59% positive). PR-AUC should be compared only on this same case and window composition.

Common montage. Both tasks use the following bipolar chains: FP1–F7–T7–P7–O1, FP2–F8–T8–P8–O2, FP1–F3–C3–P3–O1, and FP2–F4–C4–P4–O2. Adjacent pairs in each chain form the 16 channels. Older T3/T4/T5/T6 aliases are mapped to T7/T8/P7/P8. This is a 16-channel protocol; results from a 23-channel native pipeline require a separate evaluation rather than direct insertion into the main table.

Metrics. For multiclass TUEV, balanced accuracy is mean class recall and weighted F1 is weighted by test-class support. For binary CHB-MIT, PR-AUC is trapezoidal integration of the precision–recall curve. Average precision is also reported because it uses a different interpolation convention. The binary decision threshold is 0.5. Neither the threshold nor the checkpoint is selected on test labels.

B. Implementation and Training

Frontend. The 16 sinc filters share their parameters across channels. Each 201-sample symmetric FIR uses a Hamming window and zero padding of 100 samples on each side. The FFT-based Hilbert transform doubles positive-frequency coefficients, preserves the DC and Nyquist terms, and zeros negative frequencies. It is applied to the complete 5-s or 10-s filtered window. No extension beyond that window is used for the analytic transform. Responses near the boundaries may therefore depend on the finite-window convention.

The initialization is logarithmic; the learned bank is not constrained to retain constant-Q spacing or non-overlapping passbands. A token indexes a particular learned filter, rather than presupposing a physiological frequency category. Each content projection maps 50 samples to 64 coordinates. The envelope projection is affine and has a learned coordinatewise scale, absorbed into P_E and $\mathbf{b}_E$ in Eq. (2). The waveform projection has no bias and is initialized to zero. Its full trained rank in Tab. 6 confirms that it does not remain the initial zero map. Content controls preserve these projection shapes and initialization rules.

Reader. Bipolar channel coordinates concatenate the two electrode endpoints before a two-layer MLP. Dropout acts in the feed-forward sublayer. Global mean pooling precedes layer normalization and the classifier.

For C channels, B bands, and P intervals, the number of attention scores per head and block is

$$CBP^2 + BPC^2 + CPB^2 = CBP(P + C + B). \quad (3)$$

This describes attention-score complexity, not measured throughput. Projection and feed-forward costs remain proportional to token count and feature dimension. TUEV has $16 \times 16 \times 20 = 5{,}120$ tokens; CHB-MIT has $16 \times 16 \times 40 = 10{,}240$.

Training augmentation. One of five transformations is sampled uniformly per training batch: time reversal with probability 0.5, Gaussian noise with standard deviation 0.05 in stored signal units, masking 20% of temporal samples, masking 20% of channels, or masking Fourier bins with probability 0.2. Transformations act on the input before either frontend. The raw-row and band-content arms use the same choices. No training augmentation is applied during validation, testing, or token recovery.

Table 5. **TUEV test-category breakdown.** The two validation-selected checkpoints are evaluated on the same 29,421 windows. Bold marks the better tokenizer within each class and metric.

		Raw-patch rows			Demodulation-first		
Category	Support	Precision	Recall	F1	Precision	Recall	F1
SPSW	567	0.024	0.034	0.028	**0.925**	**0.109**	**0.196**
GPED	4,677	0.743	0.679	0.710	**0.915**	**0.779**	**0.842**
PLED	1,998	**0.661**	0.467	0.547	0.641	**0.792**	**0.709**
EYEM	329	0.396	**0.669**	0.497	**0.585**	0.562	**0.574**
ARTF	2,204	0.390	0.595	0.471	**0.463**	**0.637**	**0.536**
BCKG	19,646	0.900	0.871	0.885	**0.926**	**0.922**	**0.924**

Table 6. **Trained projection matrices.** Each matrix has shape 64×50. Full column rank explains local recoverability; conditioning determines sensitivity to numerical error.

Dataset	Projection	Rank	Condition no.
TUEV	Waveform	50	3.90×10^4
	Log-envelope	50	11.66
CHB-MIT	Waveform	50	3.25×10^4
	Log-envelope	50	11.64

Checkpoint selection. TUEV is validated once per epoch. CHB-MIT is validated every 200 optimizer steps and at epoch end. The patience counter counts non-improving validation evaluations; the stopping condition is checked at epoch boundaries. Patience is 20 for both tasks. The completed TUEV models finish all 20 epochs, whereas the CHB-MIT reference stops by patience after four epochs. Neither reported reference is terminated by its wall-clock budget. The selected TUEV reference checkpoint comes from evaluation 1 (zero-based), and the raw-row checkpoint from evaluation 14; the CHB-MIT reference comes from evaluation 130. All selection uses the validation metric specified in Sec. 4.

C. Recoverability of Paired Content

Why reconstruction is possible. Write a token before positional embeddings as $\mathbf{h} = [P_E\boldsymbol{\ell} + \mathbf{b}_E; P_W\mathbf{w}]$, where $\boldsymbol{\ell} = \log(1 + \mathbf{e})$. If both projection matrices have column rank L, their Moore–Penrose inverses recover the two local input vectors in exact arithmetic:

$$\boldsymbol{\ell} = P_E^{\dagger}(\mathbf{h}_E - \mathbf{b}_E), \qquad \mathbf{w} = P_W^{\dagger}\mathbf{h}_W. \tag{4}$$

Interleaving token coordinates merely changes how the two blocks are indexed. The envelope follows by applying $\exp(\boldsymbol{\ell}) - 1$. This statement concerns the band-limited waveform and its envelope; it is not a claim that the complete original recording can be inverted from the filterbank.

Readout protocol. For each dataset we select 128 training, 32 validation, and 64 test windows with fixed seed 2. Four evenly spaced channel indices, all 16 bands, and three fixed interior patch indices are used per window. The sample counts for fitting, validation, and testing are consequently 24,576, 6,144, and 12,288 tokens. Separate ridge readouts predict the 50 waveform or log-envelope samples from the 128 token coordinates. Each readout is shared across channel, band, and patch indices. The regularization coefficient is selected from $\{10^{-8}, 10^{-5}, 10^{-2}, 1\}$ by validation reconstruction error; both datasets select 10^{-8} for both targets.

Token coordinates are standardized with training-set means and standard deviations before fitting each ridge readout. Normalized RMSE is the square root of the total squared prediction error divided by the target's summed squared deviations from its per-coordinate test-set mean. For the physical envelope, obtained by exponentiating the log-envelope prediction, normalized RMSE is 1.12×10^{-6} on TUEV and 1.38×10^{-6} on CHB-MIT. A second check using the known projection weights and Eq. (4) produces the same qualitative result. Figure 2 shows a fixed example, not a reconstruction chosen to minimize error: the first held-out window and selected channel, band 12, and the middle of the three sampled intervals.

What a linear patch map cannot compute. For a nonzero filtered signal w, its envelope satisfies $|\mathcal{A}(-w)| = |\mathcal{A}(w)|$, where $\mathcal{A}$ is the linear analytic-signal operator. A linear map instead satisfies $F(-w) = -F(w)$. It therefore cannot equal the nonzero envelope map for both w and $-w$. This does not imply that a linear token loses the waveform, or that a nonlinear encoder cannot learn to compute an envelope. It identifies the computation explicitly moved ahead of the encoder in our representation.

Event categories. SPSW denotes spike and sharp waves, GPED generalized periodic epileptiform discharges, PLED periodic lateralized epileptiform discharges, EYEM eye movement, ARTF artifact, and BCKG background.